VR INTERFACES

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INPUT DEVICES

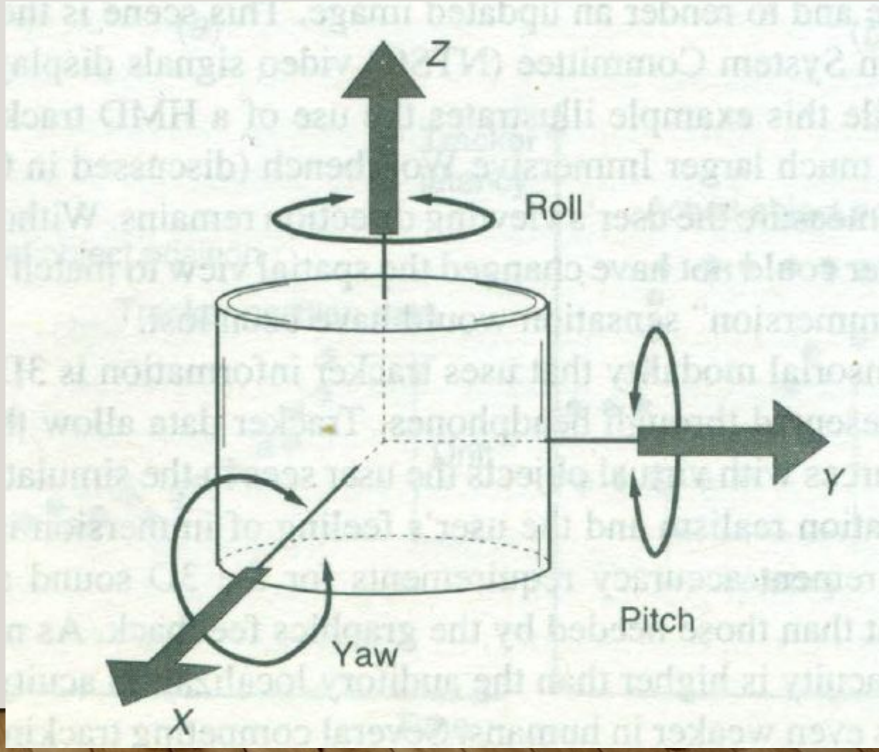
- Trackers
- VR Navigation
- Gesture Interface

3D POSITION TRACKERS

- Need is realtime position and orientation of moving object within some reference frame
- Application for this need: Robotics, Missile tracking, navigation, CAD, architecture
- Varying requirements of parameters like: measurement range, precision and temporal update rates
- Ex: GPS report position with an accuracy of 1-5m. Good for Car navigation
- VR focus on more stringent accuracy requirements

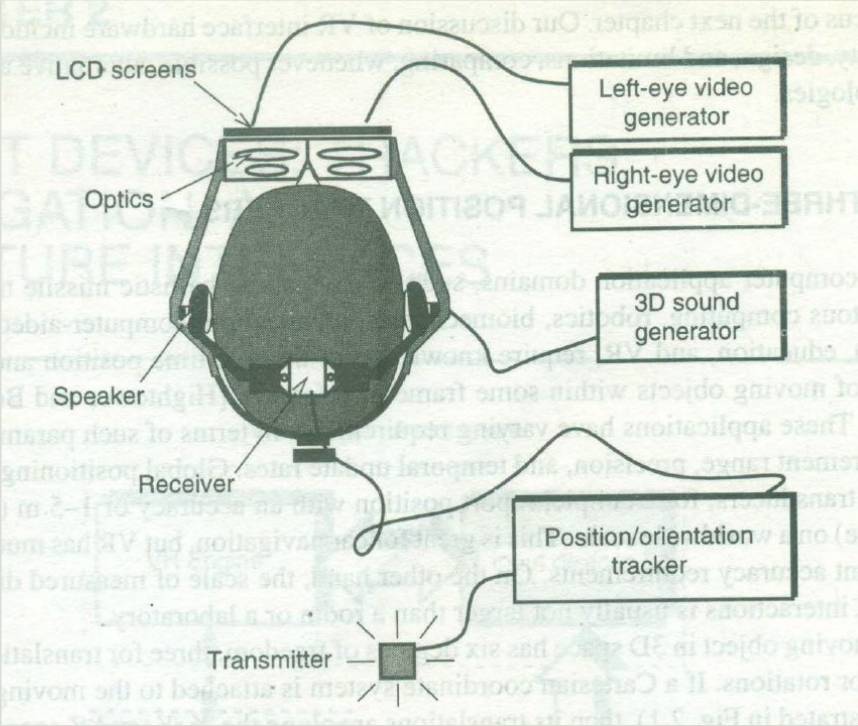
TRACKER

- Moving object in 3D has 6 degree of freedom
  - 3 translations and 3 rotations
- Objects rotation about x, y and z axis are called yaw, pitch and roll respectively
- Tracker: Special hardware to measure the position and orientation of moving object



HMD TRACKER

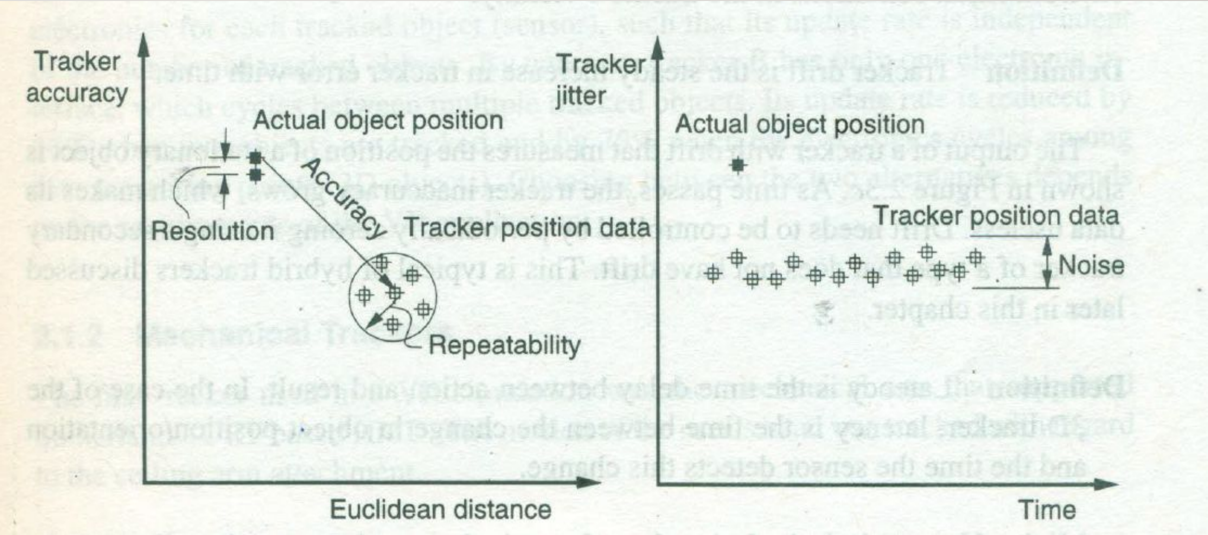
- Tracker receiver is placed on HMD
- User’s head motion is sampled and sent as electronic signal to the host computer
- 3D tracker required to change the spatial view according to head orientation : Immersion sensation achieved.
- Another sensorial modality is 3D Sound presented in Headphones
- Co-locate sound sources with virtual objects the user sees in Simulation
- Accuracy requirement for 3D sound is less than the 3D graphics

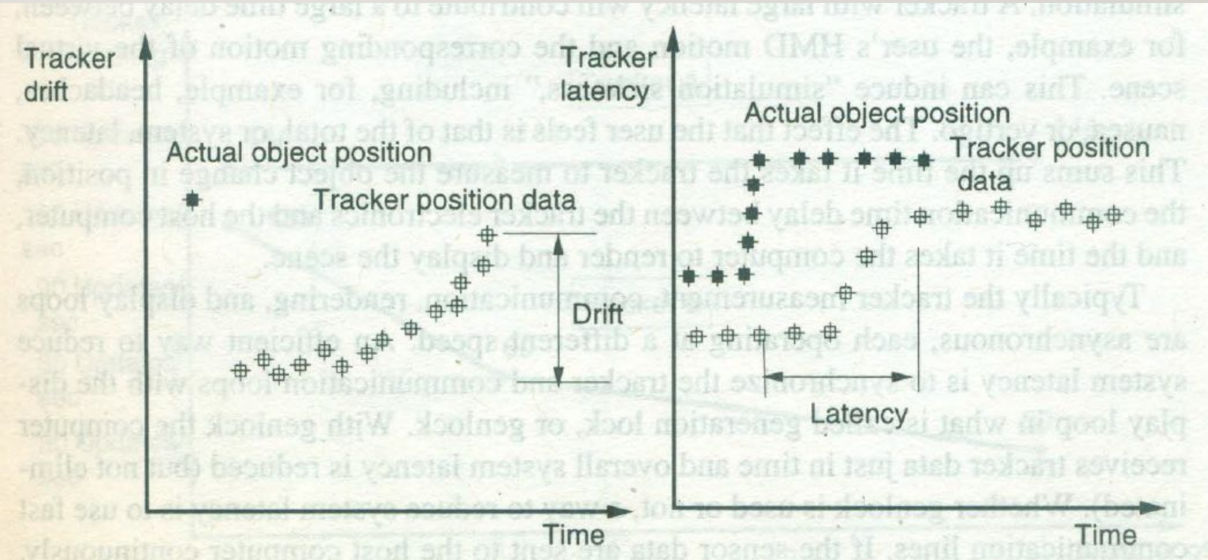


TRACKER PERFORMANCE PARAMETERS

- Accuracy (translation & rotation)
- Jitter
- Drift
- Latency

- wrt distance and time
- Smaller the difference better the performance





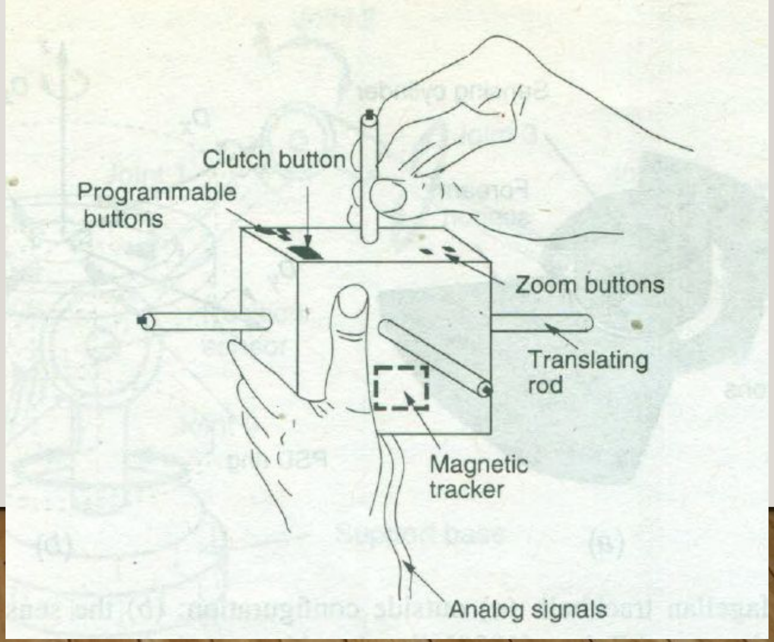
NAVIGATION AND MANIPULATION INTERFACES

- Allows interactive change of view of virtual environment and allows to explore the selected object of interest
- It can be done either through absolute coordinates or relative coordinates
- Position of VR object controlled by 6 signed entities
- Velocity is controlled by translation/rotation increments

TRACKER BASED N-M INTERFACE

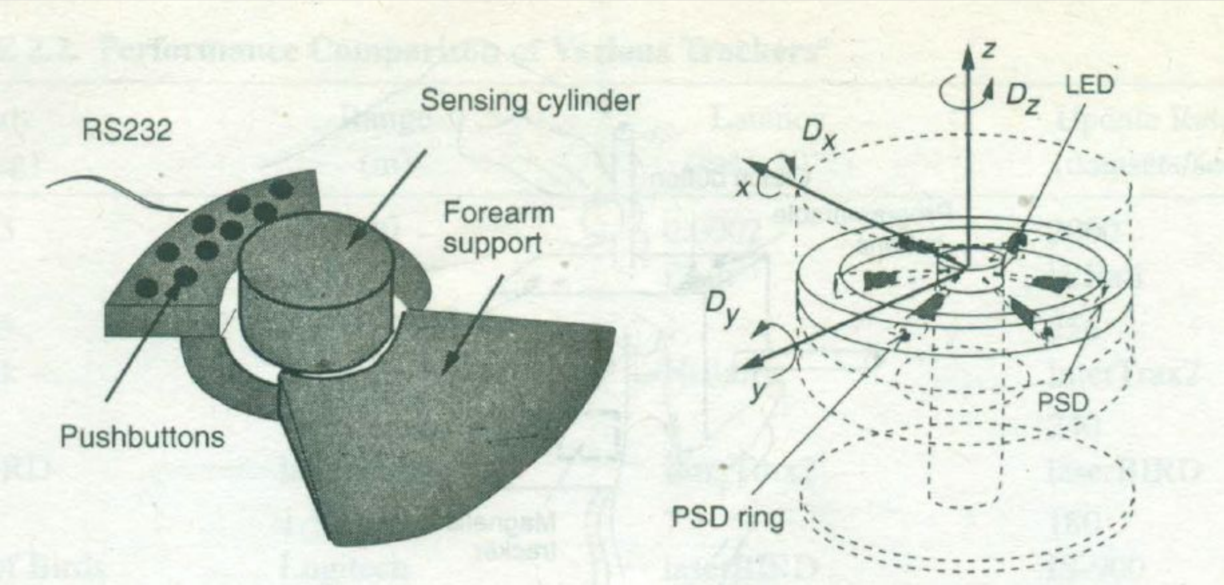
- Trackers with user programmable push buttons become navigation/manipulation interface
- Ex: 3Ball, 3D Mouse
- A virtual camera moves along 3D vector and selects the object when pushbutton is pressed
- Wand: projects a virtual ray along the 3D vector and by pushing the button the object intersected by the ray will be selected
- Cubic Mouse: 3 translating rods, 6 pushbuttons
- Designed for dual hand operation
- Hold the mouse with non dominant hand and push the rods

and buttons using dominant hand



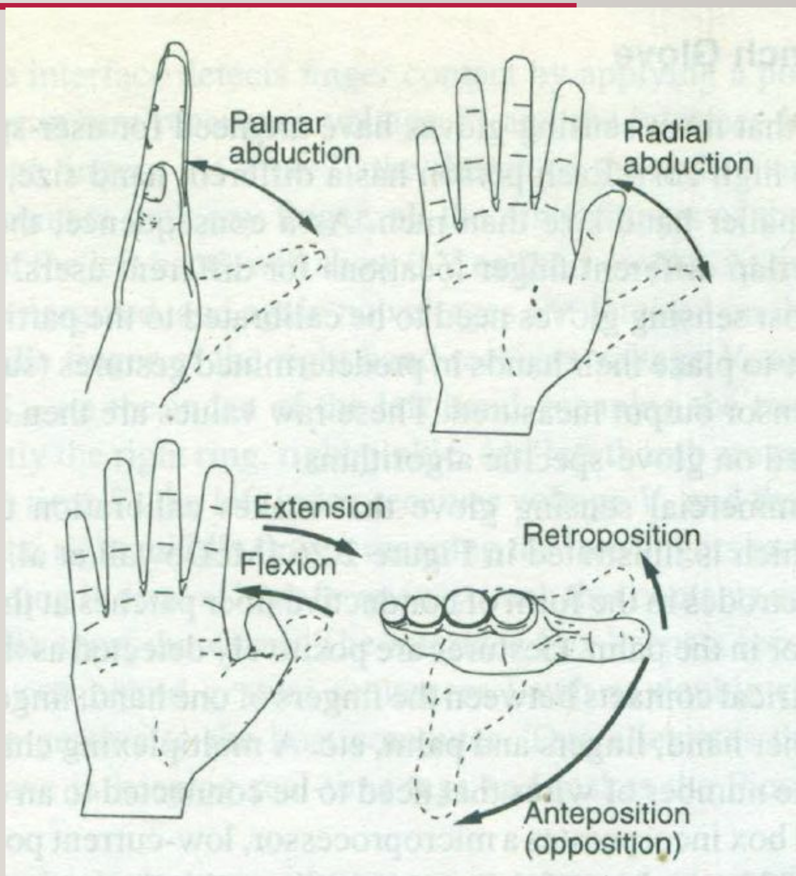
TRACKBALLS – N-M INTERFACE

- Works on relative coordinates
- Sensorized cylinder that measures three forces and three torques
- Force and torque are measured using Spring Deformation Law
- Center is fixed with 6 LEDs
- Moving external cylinders with 6 photo sensors
- Force and torques are applied on moving shells
- Photo sensor output is used to measure
- Measured value is sent to host using RS232 serial line

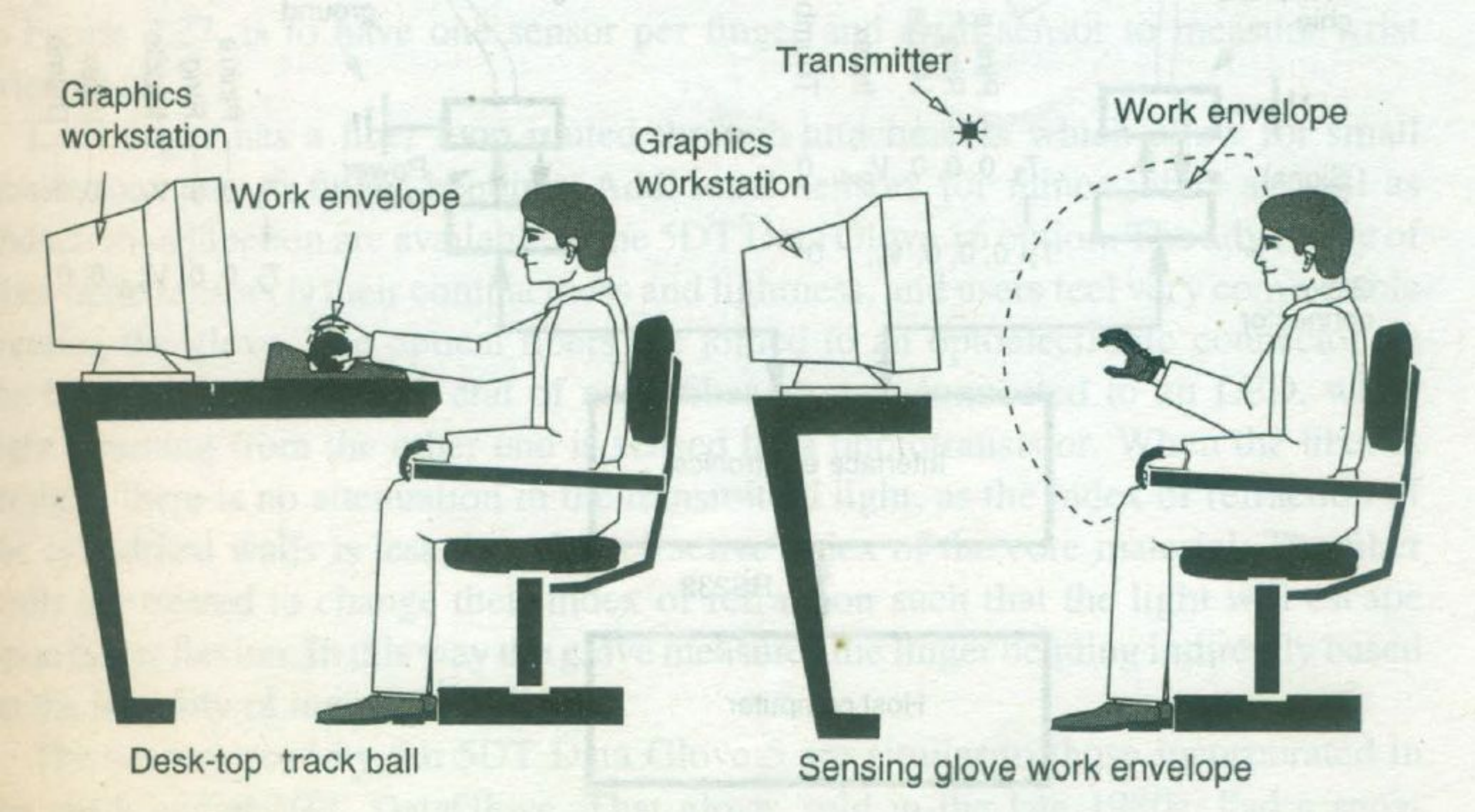


GESTURE INTERFACES

- Other interfaces limitation: limited to the freedom of users hand to a small area close to desk
- Large gesture based interactions, I/O devices need to provide free movement of hand within a certain volume.
- It is also desirable to allow sensing of individual finger motions
- Ex: Sensing Gloves that have embedded sensors to measure finger movements.
- Parameters involved are: type of sensors, number of sensors, sensor resolution, glove sampling rate, tethered or wireless



- Resulting sense glove work envelop is much larger than that of joysticks or trackballs



OUTPUT DEVICES

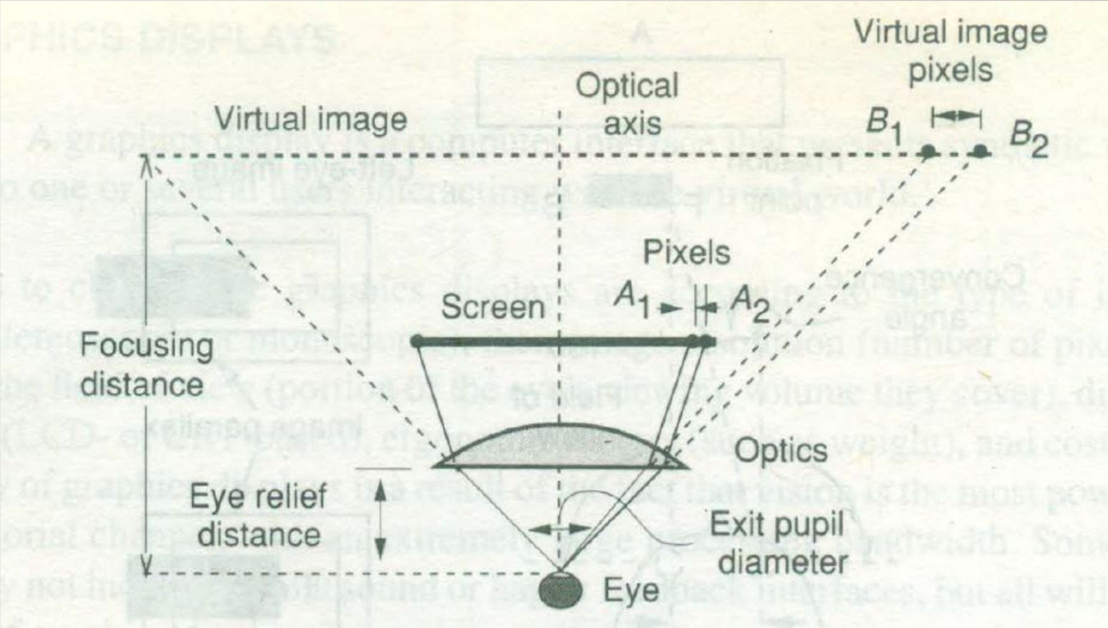
- 3D Sound
- Haptic Display

PERSONAL GRAPHICS DISPLAY

- Graphics display that outputs a virtual scene for a single user
- It can be: monoscopic or stereoscopic, monocular (for single eye) or binocular
- Further grouped as
  - Head mounted displays (HMD)
  - Hand supported displays
  - Floor supported displays
  - Auto stereoscopic displays

HEAD MOUNTED DISPLAYS (HMD)

- Special optics placed between image panel and user's eyes to reduce the strain on human eyes
- Optics also magnifies the image to fill the FOV
- Hence granularity of HMD becomes important
- Display technology can be LCDs or CRT (as they provide higher resolution). Accept NTSC/PAL signals
- VR requires RGB to NTSC/PAL conversion module
- Weight, comfort and cost are additional criteria

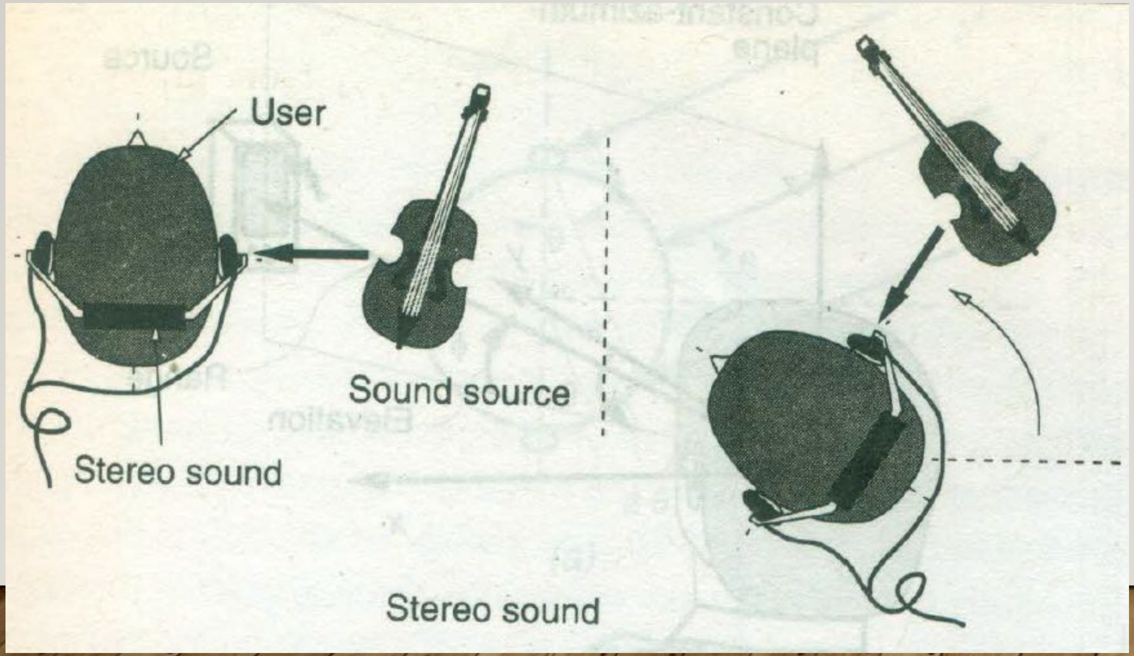


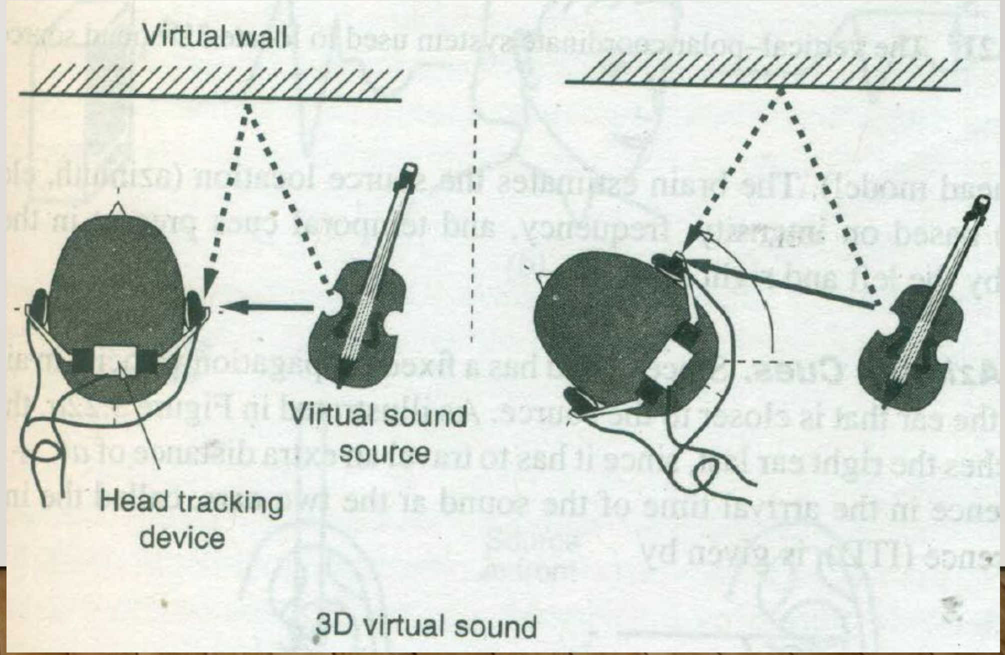
SOUND DISPLAYS

- Synthetic sound feedback to users interacting with virtual world
- It can be monoaural (same sound for both ears) or binaural (different sound for both ears)
- Ex: User looking at virtual ball bouncing in virtual room, then it can be monoaural is sufficient if the ball is always in front. But if the user is wearing HMD and the ball can go out of FOV then binaural is required
- By adding sound, user's interactivity, immersion and perceived image quality increases

DIFFERENCE BETWEEN STEREO SOUND AND 3D SOUND

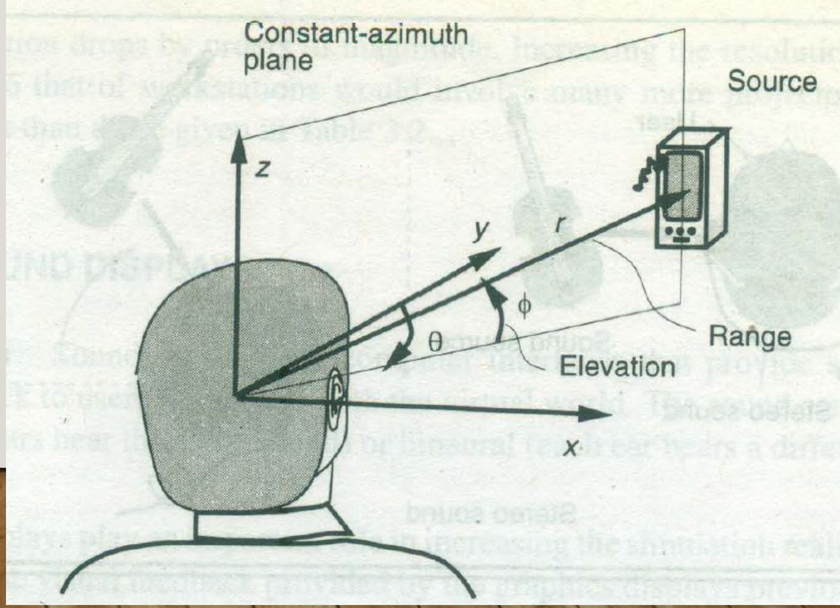
- Stereo sound is not external. When the user turns left, violin will play in the left
- 3D sound contains psychoacoustic information, the sound is synthesized





VERTICAL POLAR COORDINATE SYSTEM

- In Human Auditory System
- Sound source location is determined by 3 variables: azimuth, elevation and range
- Azimuth: angle between nose and a plane containing source and vertical z axis
- Elevation is the angle made by the horizontal plane passing through source and center of head
- Range is the distance to source along this line



HAPTIC FEEDBACK

- Greek term hapthai mean touch
- They convey important sensorial information that helps achieve tactile identification of virtual objects
- It becomes mandatory whenever visual feedback cannot be generated like when the object is occluded or in dark environment
- The feedback is limited to hand as it has highest touch receptors
- Touch and force, two modalities are considered
- Touch feedback gives surface geometry, roughness, slippage and temperature
- Force feedback gives surface compliance, object weight and inertia

CHALLENGES OF HAPTIC FEEDBACK

- User safety and comfort
  - User interacting with virtual objects but the force he feels is real
  - Good design is to fail safe so that users are not subjected to accidents if system fails
- Potability and user comfort
  - If it is too heavy and bulky user will get tired soon
  - It also refers to ease of use and installation
- Complexity and cost
- Self contained : should not require additional wiring and piping